

# Inline and Display Mathematics

Source-backed grouped formula sample

## 1 Expressions

**Expression 1.** The following expression is taken from a source-backed formula corpus.

$$\mathcal{L} = \frac{1}{2}(\tilde{\Psi}^*(i\tilde{\partial}_t - \tilde{A}_0)\tilde{\Psi} + c.c.) + \tilde{A}_0 - \frac{1}{2}|\tilde{D}_i\tilde{\Psi}|^2 + \frac{1}{2}(\frac{1}{\beta}\tilde{\mathbf{E}}^2 - \tilde{\mathbf{B}}^2) - \frac{1}{8}\kappa^2(\tilde{\Psi}\tilde{\Psi}^* - 1)^2$$

**Expression 2.** The following expression is taken from a source-backed formula corpus.

$$Hom_{[R/G^\sigma](\mathbb{F}_q)}((r',\varepsilon'),(r'',\varepsilon''))=\delta_{\varepsilon',\varepsilon''}\mathrm{Trans}_{G_e^{\sigma,\varepsilon'}(\mathbb{F}_q)}(r',r'').$$

**Expression 3.** The following expression is taken from a source-backed formula corpus.

$$T_{rh} \approx \left(\frac{90}{\pi^2 g_*}\right)^{\frac{1}{4}} \min\left(\sqrt{H(\phi_e)}, \sqrt{\Gamma}\right) \approx \left(\frac{30}{\pi^2 g_*}\right)^{\frac{1}{4}} \min\left(\Delta, \left(\frac{3}{8\pi^2}\right)^{\frac{1}{4}} p^{\frac{3}{2}} \kappa^{\frac{5}{2p}} \Delta^{3-\frac{5q}{2p}}\right)$$

**Expression 4.** The following expression is taken from a source-backed formula corpus.

$$[A_{\mathbf{G}l_i}^\gamma(\mathbf{r})\frac{\lambda^\gamma}{2}]=V_{\mathcal{C}}(\mathbf{r})\left[A_i^\gamma(\mathbf{r})\frac{\lambda^\gamma}{2}\right]V_{\mathcal{C}}^{-1}(\mathbf{r})+\frac{i}{g}V_{\mathcal{C}}(\mathbf{r})\partial_iV_{\mathcal{C}}^{-1}(\mathbf{r})\,,$$

**Expression 5.** The following expression is taken from a source-backed formula corpus.

$$\Delta_i^{KK} = \frac{\bar{b}_i}{2\pi} \sum_{(m_1,m_2) \neq (0,0)} \ln \frac{M_s}{M_{m_1,m_2}} - \frac{\bar{b}_i}{4\pi} \sum_{(m_1,m_2) \neq (0,0)} \int_{-1/2}^{1/2} d\tau_1 \int_0^{\kappa_{\vec{m}} \sqrt{1-\tau_1^2}} \frac{dt}{t} (e^{-t} - 1),$$